



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Queue Using Array

Presented by

M.Kyathi Sai Prya(192572101)

Shaik Ayesha Tabassum(192525074)

G.Hima Bindu(192511377)

INTRODUCTION

What is Queue?

- A **Queue** is a linear data structure.
- It follows the **FIFO (First In, First Out)** principle.
- The first element inserted is the first element removed.
- Queue can be implemented using an array or linked list.
- Unlike a stack, where insertion and deletion occur from the same end, a queue performs insertion at the **rear** and deletion from the **front**.

Queue Representation Using Array

- Index

0 1 2 3 4

+-----+-----+-----+-----+-----+

| 10 | 20 | 30 | | |

+-----+-----+-----+-----+-----+

- Front = 0
- Rear = 2

Explanation

- Array stores all queue elements.
- **Front** points to the first element.
- **Rear** points to the last inserted element.
- Maximum number of elements depends on the array size.

Queue Operations

- **Enqueue** – Insert an element at the rear.
- **Dequeue** – Delete an element from the front.
- **Display** – Show all queue elements.
- **Peek** – Display the front element.

1. In a deque (double-ended queue) with elements [5,10,15,20] if an operation is performed to insert the element '25' at the front and then delete an element from the rear, what will be the rear element of the deque?

- A.15 B.20 C.25 D.10

2. A linear queue has size 5. The queue contains: [_, _, 30, 40, 50] where front = 2 and rear = 4. What happens when Enqueue(60) is performed?

- A.60 is inserted at index 0
B.60 is inserted at index 1
C.Queue Overflow
D.60 replaces 30

3. A priority queue Q is used to implement a stack that stores characters. $PUSH(C)$ is implemented as $INSERT(Q, C, K)$ where K is an appropriate integer key chosen by the implementation. POP is implemented as $DELETEMIN(Q)$. For a sequence of operations, the keys chosen are in

- A. non-increasing order
- B. non-decreasing order
- C. strictly increasing order
- D. strictly decreasing order

4. Five people P, Q, R, S and T are standing in a queue. R is standing between P and T . P is just behind Q and Q is second in the queue. Who is second last in the queue?

- A. T
- B. S
- C. R
- D. P

5.How many stacks are required to implement a queue? Where no other data structure like arrays or linked lists is available.

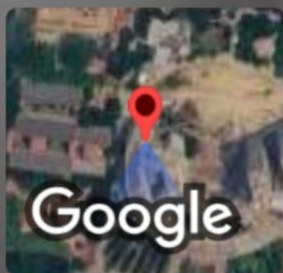
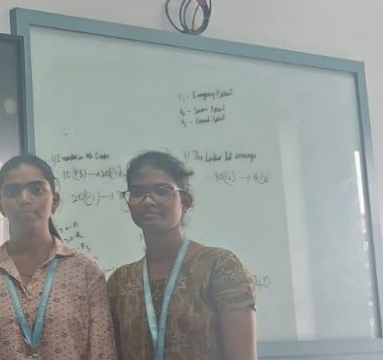
A.1

B.2

C.3

D.4

THANK YOU..



Mevalurkuppam, Tamil Nadu, India 🇮🇳

Eee Department, Mevalurkuppam, Tamil Nadu 602105, India

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